

Nand2Tetris

Project 1 and Project 2: HDL Implementation Report

Boolean Logic and Boolean Arithmetic

Course: Nand2Tetris

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Table of Contents

Project 1	3
Not	3
And	4
Or	5
Xor	6
Mux	7
DMux	8
Not16	9
And16	10
Or16	11
Mux16	12
Or8Way	13
Mux4Way16	14
Mux8Way16	15
DMux4Way	16
DMux8Way	17
Project 2	18
HalfAdder	18
FullAdder	19
Add16	20
ALU	21
Conclusion	22

Project 1

Not

NAND2Tetris / Hardware Simulator / Not.hdl

HDL ☒ --LBSt Project 1 Not

Chip Not Eval Reset Clock: 0

Test Steps: 5 Outputs: 3

Slow Fast

Test Script Compare File Output File Diff Table

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.hdl
5 /**
6  * Not gate:
7  * if (in) out = 0, else out = 1
8  */
9 CHIP Not {
10     IN in;
11     OUT out;
12
13     PARTS:
14     Nand (a=in , b = in, out=out);
15
16     //// Replace this comment with your code.
17 }
```

Input pins
in 1

Output pins
out 0

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.tst
5
6 load Not.hdl,
7 compare-to Not.cmp,
8 output-list in out;
9
10 set in 0,
11 eval,
12 output;
13
14 set in 1,
15 eval,
16 output;
```

Simulation successful: The output file is identical to the compare file

And

NAND2Tetris / Hardware Simulator / And.hdl

HDL

--LBSt

Project 1

And

Chip And

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/And.hdl

5 /**

6 * And gate:

7 * if (a and b) out = 1, else out = 0

8 */

9 CHIP And {

10 IN a, b;

11 OUT out;

12

13 PARTS:

14 Nand (a=a , b = b , out=nandOut);

15 Not(in=nandOut, out=out);

16 //// Replace this comment with your code.

17 }

Input pins

a 0

b 0

Output pins

out 0

Internal pins

nandOut 1

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/And.tst

5

6 load And.hdl,

7 compare-to And.cmp,

8 output-list a b out;

9

10 set a 0,

11 set b 0,

12 eval,

13 output;

14

15 set a 0,

16 set b 1,

17 eval,

18 output;

19

20 set a 1,

21 set b 0,

22 eval,

23 output;

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

HDL code: No syntax errors

Or

NAND2Tetris / Hardware Simulator / Or.hdl

HDL

--LBST

Project 1

Or

Chip Or

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or.hdl

5 /**

6 * Or gate:

7 * if (a or b) out = 1, else out = 0

8 */

9 CHIP Or {

10 IN a, b;

11 OUT out;

12

13 PARTS:

14 Not(in=a, out=nota);

15 Not(in=b, out=notb);

16 Nand(a=nota, b =notb , out=out);

17

18 //// Replace this comment with your code.

19 }

Input pins

a0

b0

Output pins

out0

Internal pins

nota1

notb1

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or.tst

5

6 load Or.hdl,

7 compare-to Or.cmp,

8 output-list a b out;

9

10 set a 0,

11 set b 0,

12 eval,

13 output;

14

15 set a 1,

16 set b 1,

17 eval,

18 output;

19

20 set a 1,

21 set b 0,

22 eval,

23 output;

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

HDL code: No syntax errors

Xor

NAND2Tetris / Hardware Simulator / Xor.hdl

HDL

--LBSt

Project 1

Xor

1 // This file is part of the book "The Elements of Computing Systems"

2 // by Nisan and Schocken, MIT Press.

3 // File name: projects/1/Xor.hdl

4 /**

5 * Exclusive-or gate:

6 * if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0

7 */

8 CHIP Xor {

9 IN a, b;

10 OUT out;

11

12 PARTS:

13 Not(in=a,out=notA);

14 Not(in=b,out=notB);

15 And (a=a,b=notB, out=w1);

16 And (a=notA, b=b , out=w2);

17 Or(a=w1, b=w2, out=out);

18 //// Replace this comment with your code.

19 }

20 }

Chip Xor

Eval Reset Clock: 0

Test

Steps: 5 Outputs: 0

Slow Fast

Test Script Compare File Output File Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Xor.tst

5

6 load Xor.hdl,

7 compare-to Xor.cmp,

8 output-list a b out;

9

10 set a 0,

11 set b 0,

12 eval,

13 output;

14

15 set a 1,

16 set b 0,

17 eval,

18 output;

19

20 set a 1,

21 set b 1,

22 eval,

23 output;

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

Open an HDL file using this menu

Input pins

a 0

b 0

Output pins

out 0

Internal pins

notA 1

notB 1

w1 0

w2 0

HDL code: No syntax errors

Mux

NAND2Tetris / Hardware Simulator / Mux.hdl

HDL

--LBSt

Project 1

Mux

Chip Mux

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Mux.hdl

5 /**

6 * Multiplexor:

7 * if (sel = 0) out = a, else out = b

8 */

9 CHIP Mux {

10 IN a, b, sel;

11 OUT out;

12

13 PARTS:

14 Not(in=sel,out=notSel);

15 And(a=a, b = notSel, out=w1);

16 And(a=b, b=sel, out=w2);

17 Or(a=w1, b=w2 , out=out);

18 //// Replace this comment with your code.

19 }

Input pins

a 0

b 0

sel 0

Output pins

out 0

Internal pins

notSel 1

w1 0

w2 0

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Mux.tst

5

6 load Mux.hdl,

7 compare-to Mux.cmp,

8 output-list a b sel out;

9

10 set a 0,

11 set b 0,

12 set sel 0,

13 eval,

14 output;

15

16 set sel 1,

17 eval,

18 output;

19

20 set a 0,

21 set b 1,

22 set sel 0,

23 eval,

24 output;

25

26 set sel 1,

27 eval,

28 output;

29

30 set a 1,

31 set b 0,

32 set sel 0,

33 eval,

34 output;

DMux

NAND2Tetris / Hardware Simulator / DMux.hdl

HDL

--LBSt

Project 1

DMux

Chip DMux

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux.hdl

5 /**

6 * Demultiplexor:

7 * [a, b] = [in, 0] if sel = 0

8 * [0, in] if sel = 1

9 */

10 CHIP DMux {

11 IN in, sel;

12 OUT a, b;

13

14 PARTS:

15 Not(in = sel, out=nsel);

16 And(a=in, b=nsel, out=a);

17 And(a=in, b=sel, out=b);

18

19 //// Replace this comment with your code.

20 }

Input pins

in 0

sel 0

Output pins

a 0

b 0

Internal pins

nsel 1

Test

Steps: 5

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux.tst

5

6 load DMux.hdl,

7 compare-to DMux.cmp,

8 output-list in sel a b;

9

10 set in 0,

11 set sel 0,

12 eval,

13 output;

14

15 set sel 1,

16 eval,

17 output;

18

19 set in 1,

20 set sel 0,

21 eval,

22 output;

23

24 set sel 1,

25 eval,

26 output;

HDL code: No syntax errors

Not16

NAND2Tetris / Hardware Simulator / Not16.hdl

HDL

--LBSt

Project 1

Not16

Chip Not16

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 6

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/01/Not16.hdl
5 /**
6  * 16-bit Not gate:
7  * for i = 0, ..., 15:
8  * out[i] = Not(a[i])
9  */
10 CHIP Not16 {
11     IN in[16];
12     OUT out[16];
13
14     PARTS:
15     Not(in = in[0], out=out[0]);
16     Not(in = in[1], out=out[1]);
17     Not(in = in[2], out=out[2]);
18     Not(in = in[3], out=out[3]);
19     Not(in = in[4], out=out[4]);
20     Not(in = in[5], out=out[5]);
21     Not(in = in[6], out=out[6]);
22     Not(in = in[7], out=out[7]);
23     Not(in = in[8], out=out[8]);
24     Not(in = in[9], out=out[9]);
25     Not(in = in[10], out=out[10]);
26     Not(in = in[11], out=out[11]);
27     Not(in = in[12], out=out[12]);
28     Not(in = in[13], out=out[13]);
29     Not(in = in[14], out=out[14]);
30     Not(in = in[15], out=out[15]);
31
32     //// Replace this comment with your code.
33 }
34 }
```

Input pins

in 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0

dec

Output pins

out 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1

dec

Test Script

Compare File

Output File

Diff Table

```
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not16.tst
5
6 load Not16.hdl,
7 compare-to Not16.cmp,
8 output-list in%B1.16.1 out%B1.16.1;
9
10 set in %B0000000000000000,
11 eval,
12 output;
13
14 set in %B1111111111111111,
15 eval,
16 output;
17
18 set in %B1010101010101010,
19 eval,
20 output;
21
22 set in %B0011110011000011,
23 eval,
24 output;
25
26 set in %B0001001000110100,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

And16

NAND2Tetris / Hardware Simulator / And16.hdl

HDL

--LBSt

Project 1

And16

Chip And16

Eval

Reset

Clock: 0

Test

Steps: 8

Outputs: 0

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And16.hdl
5 /**
6  * 16-bit And gate:
7  * for i = 0, ..., 15:
8  * out[i] = a[i] And b[i]
9  */
10 CHIP And16 {
11     IN a[16], b[16];
12     OUT out[16];
13
14     PARTS:
15     And(a=a[0], b=b[0], out=out[0]);
16     And(a=a[1], b=b[1], out=out[1]);
17     And(a=a[2], b=b[2], out=out[2]);
18     And(a=a[3], b=b[3], out=out[3]);
19     And(a=a[4], b=b[4], out=out[4]);
20     And(a=a[5], b=b[5], out=out[5]);
21     And(a=a[6], b=b[6], out=out[6]);
22     And(a=a[7], b=b[7], out=out[7]);
23     And(a=a[8], b=b[8], out=out[8]);
24     And(a=a[9], b=b[9], out=out[9]);
25     And(a=a[10], b=b[10], out=out[10]);
26     And(a=a[11], b=b[11], out=out[11]);
27     And(a=a[12], b=b[12], out=out[12]);
28     And(a=a[13], b=b[13], out=out[13]);
29     And(a=a[14], b=b[14], out=out[14]);
30     And(a=a[15], b=b[15], out=out[15]);
31
32
33     //// Replace this comment with your code.
34 }

```

Input pins

a 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

b 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Test Script

Compare File

Output File

Diff Table

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And16.tst
5
6 load And16.hdl,
7 compare-to And16.cmp,
8 output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
9
10 set a %B0000000000000000,
11 set b %B0000000000000000,
12 eval,
13 output;
14
15 set a %B0000000000000000,
16 set b %B1111111111111111,
17 eval,
18 output;
19
20 set a %B1111111111111111,
21 set b %B1111111111111111,
22 eval,
23 output;
24
25 set a %B1010101010101010,
26 set b %B0101010101010101,
27 eval,
28 output;
29
30 set a %B0011110011000011,
31 set b %B0000111111110000,
32 eval,
33 output;
34

```

HDL code: No syntax errors

Or16

NAND2Tetris / Hardware Simulator / Or16.hdl

HDL

--LBSt

Project 1

Or16

Chip Or16

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or16.hdl

5 /**

6 * 16-bit Or gate:

7 * for i = 0, ..., 15:

8 * out[i] = a[i] Or b[i]

9 */

10 CHIP Or16 {

11 IN a[16], b[16];

12 OUT out[16];

13

14 PARTS:

15 Or(a=a[0], b=b[0], out=out[0]);

16 Or(a=a[1], b=b[1], out=out[1]);

17 Or(a=a[2], b=b[2], out=out[2]);

18 Or(a=a[3], b=b[3], out=out[3]);

19 Or(a=a[4], b=b[4], out=out[4]);

20 Or(a=a[5], b=b[5], out=out[5]);

21 Or(a=a[6], b=b[6], out=out[6]);

22 Or(a=a[7], b=b[7], out=out[7]);

23 Or(a=a[8], b=b[8], out=out[8]);

24 Or(a=a[9], b=b[9], out=out[9]);

25 Or(a=a[10], b=b[10], out=out[10]);

26 Or(a=a[11], b=b[11], out=out[11]);

27 Or(a=a[12], b=b[12], out=out[12]);

28 Or(a=a[13], b=b[13], out=out[13]);

29 Or(a=a[14], b=b[14], out=out[14]);

30 Or(a=a[15], b=b[15], out=out[15]);

31

32 //// Replace this comment with your code.

33 }

Input pins

a 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

b 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Test

Steps: 8

Outputs: 0

Slow Fast

Test Script Compare File Output File Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or16.tst

5

6 load Or16.hdl,

7 compare-to Or16.cmp,

8 output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;

9

10 set a %B0000000000000000,

11 set b %B0000000000000000,

12 eval,

13 output;

14

15 set a %B0000000000000000,

16 set b %B1111111111111111,

17 eval,

18 output;

19

20 set a %B1111111111111111,

21 set b %B1111111111111111,

22 eval,

23 output;

24

25 set a %B1010101010101010,

26 set b %B0101010101010101,

27 eval,

28 output;

29

30 set a %B0011110011000011,

31 set b %B0000111111110000,

32 eval,

33 output;

34

HDL code: No syntax errors

Mux16

NAND2Tetris / Hardware Simulator / Mux16.hdl

HDL

--LBSt

Project 1

Mux16

Chip Mux16

Eval

Reset

Clock: 0

Test

Steps: 8

Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux16.hdl
5 /**
6  * 16-bit multiplexor:
7  * for i = 0, ..., 15:
8  * if (sel = 0) out[i] = a[i], else out[i] = b[i]
9  */
10 CHIP Mux16 {
11     IN a[16], b[16], sel;
12     OUT out[16];
13
14     PARTS:
15     Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
16     Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
17     Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
18     Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
19     Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
20     Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
21     Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
22     Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
23     Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
24     Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
25     Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
26     Mux(a=a[11], b=b[11], sel=sel, out=out[11]);
27     Mux(a=a[12], b=b[12], sel=sel, out=out[12]);
28     Mux(a=a[13], b=b[13], sel=sel, out=out[13]);
29     Mux(a=a[14], b=b[14], sel=sel, out=out[14]);
30     Mux(a=a[15], b=b[15], sel=sel, out=out[15]);
31
32
33
34     //// Replace this comment with your code.
35 }

```

Input pins

a

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

b

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

sel

0

Output pins

out

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux16.tst
5
6 load Mux16.hdl,
7 compare-to Mux16.cmp,
8 output-list a%B1.16.1 b%B1.16.1 sel out%B1.16.1;
9
10 set a 0,
11 set b 0,
12 set sel 0,
13 eval,
14 output;
15
16 set sel 1,
17 eval,
18 output;
19
20 set a %B0000000000000000,
21 set b %B0001001000110100,
22 set sel 0,
23 eval,
24 output;
25
26 set sel 1,
27 eval,
28 output;
29
30 set a %B1001100001110110,
31 set b %B0000000000000000,
32 set sel 0,
33 eval,
34 output;

```

HDL code: No syntax errors

Or8Way

NAND2Tetris / Hardware Simulator / Or8Way.hdl

HDL

●

--LBSt

Project 1

Or8Way

⬇

⬇

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or8Way.hdl

5 /**

6 * 8-way Or gate:

7 * out = in[0] Or in[1] Or ... Or in[7]

8 */

9 CHIP Or8Way {

10 IN in[8];

11 OUT out;

12

13 PARTS:

14 Or(a=in[0], b= in[1], out = w1);

15 Or(a=w1 , b=in[2], out=w2);

16 Or(a=w2 , b=in[3], out=w3);

17 Or(a=w3 , b=in[4], out=w4);

18 Or(a=w4 , b=in[5], out=w5);

19 Or(a=w5 , b=in[6], out=w6);

20 Or(a=w6 , b=in[7], out=out);

21

22 //// Replace this comment with your code.

23 }

Chip Or8Way

Eval Reset Clock: 0

Input pins

in 0 0 0 0 0 0 0 0 dec

Output pins

out 0

Internal pins

w1 0

w2 0

w3 0

w4 0

w5 0

w6 0

Test

Steps: 8 Outputs: 0

Slow Fast

Test Script Compare File Output File Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Or8Way.tst

5

6 load Or8Way.hdl,

7 compare-to Or8Way.cmp,

8 output-list in%B2.8.2 out;

9

10 set in %B00000000,

11 eval,

12 output;

13

14 set in %B11111111,

15 eval,

16 output;

17

18 set in %B00010000,

19 eval,

20 output;

21

22 set in %B00000001,

23 eval,

24 output;

25

26 set in %B00100110,

27 eval,

28 output;

HDL code: No syntax errors

Mux4Way16

NAND2Tetris / Hardware Simulator / Mux4Way16.hdl

HDL

--LBSt

Project 1

Mux4Way16

Chip Mux4Way16

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Mux4Way16.hdl

5 /**

6 * 4-way 16-bit multiplexor:

7 * out = a if sel = 00

8 * b if sel = 01

9 * c if sel = 10

10 * d if sel = 11

11 */

12 CHIP Mux4Way16 {

13 IN a[16], b[16], c[16], d[16], sel[2];

14 OUT out[16];

15

16 PARTS:

17 Mux16(a=a, b=b , sel=sel[0], out=w1);

18 Mux16(a=c , b=d , sel=sel[0] , out=w2);

19 Mux16(a=w1, b=w2 , sel=sel[1], out=out);

20 //// Replace this comment with your code.

21 }

Input pins

a 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

b 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

c 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

d 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

sel 0 0 dec

Output pins

out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Internal pins

w1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

w2 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/Mux4Way16.tst

5

6 load Mux4Way16.hdl,

7 compare-to Mux4Way16.cmp,

8 output-list a%B1.16.1 b%B1.16.1 c%B1.16.1 d%B1.16.1 sel%

9

10 set a 0,

11 set b 0,

12 set c 0,

13 set d 0,

14 set sel 0,

15 eval,

16 output;

17

18 set sel 1,

19 eval,

20 output;

21

22 set sel 2,

23 eval,

24 output;

25

26 set sel 3,

27 eval,

28 output;

29

30 set a %B0001001000110100,

31 set b %B1001100001110110,

32 set c %B1010101010101010,

33 set d %B0101010101010101,

34 set sel 0,

HDL code: No syntax errors

Mux8Way16

NAND2Tetris / Hardware Simulator / Mux8Way16.hdl

HDL

--LBST

Project 1

Mux8Way16

Chip Mux8Way16

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 0

Slow Fast

Test Script

Compare File

Output File

Diff Table

```

1 This file is part of www.nand2tetris.org
2 and the book "The Elements of Computing Systems"
3 by Nisan and Schocken, MIT Press.
4 File name: projects/1/Mux8Way16.hdl
5
6 8-way 16-bit multiplexor:
7 out = a if sel = 000
8   b if sel = 001
9   c if sel = 010
10  d if sel = 011
11  e if sel = 100
12  f if sel = 101
13  g if sel = 110
14  h if sel = 111
15 /
16 IP Mux8Way16 {
17   IN a[16], b[16], c[16], d[16],
18     e[16], f[16], g[16], h[16],
19     sel[3];
20   OUT out[16];
21
22   PARTS:
23     Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], out=abcd);
24     Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], out=efgh);
25     Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
26
27     //// Replace this comment with your code.
28

```

Input pins

a

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

b

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

c

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

d

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

e

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

f

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

g

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

h

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

sel

0 0 0

dec

Output pins

out

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

Internal pins

abcd

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

efgh

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux8Way16.tst
5
6 load Mux8Way16.hdl,
7 compare-to Mux8Way16.cmp,
8 output-list a%B1.16.1 b%B1.16.1 c%B1.16.1 d%B1.16.1 e%B1
9
10 set a 0,
11 set b 0,
12 set c 0,
13 set d 0,
14 set e 0,
15 set f 0,
16 set g 0,
17 set h 0,
18 set sel 0,
19 eval,
20 output;
21
22 set sel 1,
23 eval,
24 output;
25
26 set sel 2,
27 eval,
28 output;
29
30 set sel 3,
31 eval,
32 output;
33
34 set sel 4,

```

HDL code: No syntax errors

DMux4Way

NAND2Tetris / Hardware Simulator / DMux4Way.hdl

HDL

--LBSt

Project 1

DMux4Way

Chip DMux4Way

Eval

Reset

Clock: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux4Way.hdl

5 /**

6 * 4-way demultiplexor:

7 * [a, b, c, d] = [in, 0, 0, 0] if sel = 00

8 * [0, in, 0, 0] if sel = 01

9 * [0, 0, in, 0] if sel = 10

10 * [0, 0, 0, in] if sel = 11

11 */

12 CHIP DMux4Way {

13 IN in, sel[2];

14 OUT a, b, c, d;

15

16 PARTS:

17 DMux(in=in , sel=sel[1], a=ab, b=cd);

18 DMux(in=ab, sel=sel[0], a=a, b=b);

19 DMux(in=cd, sel=sel[0], a=c , b=d);

20

21

22 //// Replace this comment with your code.

23 }

Input pins

in 0

sel 0 0 dec

Output pins

a 0

b 0

c 0

d 0

Internal pins

ab 0

cd 0

Test

Slow

Fast

Steps: 8 Outputs: 0

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux4Way.tst

5

6 load DMux4Way.hdl,

7 compare-to DMux4Way.cmp,

8 output-list in sel%B2.2.2 a b c d;

9

10 set in 0,

11 set sel %B00,

12 eval,

13 output;

14

15 set sel %B01,

16 eval,

17 output;

18

19 set sel %B10,

20 eval,

21 output;

22

23 set sel %B11,

24 eval,

25 output;

26

27 set in 1,

28 set sel %B00,

29 eval,

30 output;

31

32 set sel %B01,

33 eval,

34 output;

HDL code: No syntax errors

DMux8Way

NAND2Tetris / Hardware Simulator / DMux8Way.hdl

HDL

--LBS

Project 1

DMux8Way

Chip DMux8Way

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux8Way.hdl

5 /**

6 * 8-way demultiplexor:

7 * [a, b, c, d, e, f, g, h] = [in, 0, 0, 0, 0, 0, 0, 0]

8 * [0, in, 0, 0, 0, 0, 0, 0]

9 * [0, 0, in, 0, 0, 0, 0, 0]

10 * [0, 0, 0, in, 0, 0, 0, 0]

11 * [0, 0, 0, 0, in, 0, 0, 0]

12 * [0, 0, 0, 0, 0, in, 0, 0]

13 * [0, 0, 0, 0, 0, 0, in, 0]

14 * [0, 0, 0, 0, 0, 0, 0, in]

15 */

16 CHIP DMux8Way {

17 IN in, sel[3];

18 OUT a, b, c, d, e, f, g, h;

19

20 PARTS:

21 DMux(in=in, sel=sel[2], a=abcd, b=efgh);

22 DMux4Way(in=abcd, sel=sel[0..1], a=a, b=b, c=c, d=d);

23 DMux4Way(in=efgh, sel=sel[0..1], a=e, b=f, c=g, d=h);

24 //// Replace this comment with your code.

25 }

Input pins

in 0

sel 0 0 0 dec

Output pins

a 0

b 0

c 0

d 0

e 0

f 0

g 0

h 0

Internal pins

abcd 0

efgh 0

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/1/DMux8Way.tst

5

6 load DMux8Way.hdl,

7 compare-to DMux8Way.cmp,

8 output-list in sel%B2.3.2 a b c d e f g h;

9

10 set in 0,

11 set sel %B000,

12 eval,

13 output;

14

15 set sel %B001,

16 eval,

17 output;

18

19 set sel %B010,

20 eval,

21 output;

22

23 set sel %B011,

24 eval,

25 output;

26

27 set sel %B100,

28 eval,

29 output;

30

17

Project 2

HalfAdder

NAND2Tetris / Hardware Simulator / HalfAdder.hdl

HDL

--LBSt

Project 2

HalfAdder

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/HalfAdder.hdl

5 /**

6 * Computes the sum of two bits.

7 */

8 CHIP HalfAdder {

9 IN a, b; // 1-bit inputs

10 OUT sum, // Right bit of a + b

11 carry; // Left bit of a + b

12

13 PARTS:

14 Xor(a=a , b=b , out=sum);

15 And(a=a,b=b,out=carry);

16

17 //// Replace this comment with your code.

18 }

Chip HalfAdder

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 0

Slow Fast

Test Script Compare File Output File Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/HalfAdder.tst

5

6 load HalfAdder.hdl,

7 compare-to HalfAdder.cmp,

8 output-list a b sum carry;

9

10 set a 0,

11 set b 0,

12 eval,

13 output;

14

15 set a 0,

16 set b 1,

17 eval,

18 output;

19

20 set a 1,

21 set b 0,

22 eval,

23 output;

24

25 set a 1,

26 set b 1,

27 eval,

28 output;

HDL code: No syntax errors

FullAdder

NAND2Tetris / Hardware Simulator / FullAdder.hdl

HDL

--LBSt

Project 2

FullAdder

Chip FullAdder

Eval Reset Clock: 0

Test

Steps: 8 Outputs: 0

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/FullAdder.hdl

5 /**

6 * Computes the sum of three bits.

7 */

8 CHIP FullAdder {

9 IN a, b, c; // 1-bit inputs

10 OUT sum, // Right bit of a + b + c

11 carry; // Left bit of a + b + c

12

13 PARTS:

14 HalfAdder(a=a,b=b,sum=s1,carry=c1);

15 HalfAdder(a=s1,b=c,sum = sum,carry=c2);

16 Or(a=c1,b=c2,out=carry);

17

18

19

20 //// Replace this comment with your code.

21 }

Input pins

a 0

b 0

c 0

Output pins

sum 0

carry 0

Internal pins

s1 0

c1 0

c2 0

Test Script

Compare File

Output File

Diff Table

1 // This file is part of www.nand2tetris.org

2 // and the book "The Elements of Computing Systems"

3 // by Nisan and Schocken, MIT Press.

4 // File name: projects/2/FullAdder.tst

5

6 load FullAdder.hdl,

7 compare-to FullAdder.cmp,

8 output-list a b c sum carry%B2.1.2;

9

10 set a 0,

11 set b 0,

12 set c 0,

13 eval,

14 output;

15

16 set c 1,

17 eval,

18 output;

19

20 set b 1,

21 set c 0,

22 eval,

23 output;

24

25 set c 1,

26 eval,

27 output;

28

29 set a 1,

30 set b 0,

31 set c 0,

32 eval,

33 output;

34

HDL code: No syntax errors

Add16

NAND2Tetris / Hardware Simulator / Add16.hdl

HDL

--LBST

Project 2

Add16

Eval

Reset

Clock: 0

Test

Steps: 8

Outputs: 0

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Add16.hdl
5 /**
6  * 16-bit adder: Adds two 16-bit two's complement values
7  * The most significant carry bit is ignored.
8  */
9 CHIP Add16 {
10     IN a[16], b[16];
11     OUT out[16];
12
13     PARTS:
14     HalfAdder(a=a[0], b=b[0], sum=out[0], carry=c0);
15     FullAdder(a=a[1], b=b[1], c=c0, sum=out[1], carry=c1);
16     FullAdder(a=a[2], b=b[2], c=c1, sum=out[2], carry=c2);
17     FullAdder(a=a[3], b=b[3], c=c2, sum=out[3], carry=c3);
18     FullAdder(a=a[4], b=b[4], c=c3, sum=out[4], carry=c4);
19     FullAdder(a=a[5], b=b[5], c=c4, sum=out[5], carry=c5);
20     FullAdder(a=a[6], b=b[6], c=c5, sum=out[6], carry=c6);
21     FullAdder(a=a[7], b=b[7], c=c6, sum=out[7], carry=c7);
22     FullAdder(a=a[8], b=b[8], c=c7, sum=out[8], carry=c8);
23     FullAdder(a=a[9], b=b[9], c=c8, sum=out[9], carry=c9);
24     FullAdder(a=a[10], b=b[10], c=c9, sum=out[10], carry=c10);
25     FullAdder(a=a[11], b=b[11], c=c10, sum=out[11], carry=c11);
26     FullAdder(a=a[12], b=b[12], c=c11, sum=out[12], carry=c12);
27     FullAdder(a=a[13], b=b[13], c=c12, sum=out[13], carry=c13);
28     FullAdder(a=a[14], b=b[14], c=c13, sum=out[14], carry=c14);
29     FullAdder(a=a[15], b=b[15], c=c14, sum=out[15], carry=c15);
30
31     //// Replace this comment with your code.
32 }

```

Chip Add16

Input pins

a

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

b

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

Output pins

out

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

Internal pins

c0

0

c1

0

c2

0

c3

0

c4

0

c5

0

c6

0

c7

0

c8

0

c9

0

c10

0

c11

0

c12

0

c13

0

c14

0

c15

0

Test Script

Compare File

Output File

Diff Table

```

1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Add16.tst
5
6 load Add16.hdl,
7 compare-to Add16.cmp,
8 output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
9
10 set a %B0000000000000000,
11 set b %B0000000000000000,
12 eval,
13 output;
14
15 set a %B0000000000000000,
16 set b %B1111111111111111,
17 eval,
18 output;
19
20 set a %B1111111111111111,
21 set b %B1111111111111111,
22 eval,
23 output;
24
25 set a %B1010101010101010,
26 set b %B0101010101010101,
27 eval,
28 output;
29
30 set a %B0011110011000011,
31 set b %B0000111111110000,
32 eval,
33 output;
34

```

HDL code: No syntax errors

ALU

NAND2Tetris / Hardware Simulator / ALU.hdl

HDL

--LBSt

Project 2

ALU

Chip ALU

Eval Reset Clock: 0

Test

Steps: 41 Outputs: 0

Slow

Fast

Test Script

Compare File

Output File

Diff Table

```

29 CHIP ALU {
30   IN
31     x[16], y[16], // 16-bit inputs
32     zx, // zero the x input?
33     nx, // negate the x input?
34     zy, // zero the y input?
35     ny, // negate the y input?
36     f, // compute (out = x + y) or (out = x & y)?
37     no; // negate the out output?
38   OUT
39     out[16], // 16-bit output
40     zr, // if (out == 0) equals 1, else 0
41     ng; // if (out < 0) equals 1, else 0
42
43   PARTS:
44     Mux16(a = x, b=false, sel=zx, out=x1);
45     Not16(in=x1, out=nx1);
46     Mux16(a=x1, b=nx1, sel=nx, out =x2);
47
48     Mux16(a=y, b=false, sel=zy, out=y1);
49     Not16(in=y1, out=ny1);
50     Mux16(a=y1, b=ny1, sel=ny, out=y2);
51
52     And16(a=x2, b=y2, out = fAnd);
53     Add16(a=x2, b=y2, out=fAdd);
54     Mux16(a=fAnd, b=fAdd, sel=f, out=f0Out);
55
56     Not16(in=f0Out, out=nf0Out);
57     Mux16(a=f0Out, b=nf0Out, sel=no, out=out, out[0..7]=low8, out[8..15]=high8);
58
59     Or8Way(in=low8, out=orLow);
60     Or8Way(in=high8, out=orHigh);
61     Or(a=orLow, b=orHigh, out=notZero);
62     Not(in=notZero, out=zr);
63
64 }

```

Input pins

x

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

y

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

zx

0

nx

0

zy

0

ny

0

f

0

no

0

Output pins

out

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

zr

1

ng

0

Internal pins

x1

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

nx1

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

dec

x2

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

y1

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

ny1

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

dec

y2

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

fAnd

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

fAdd

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

dec

```

1 // This file is part of the NAND2Tetris project.
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/ALU.tst
5
6 load ALU.hdl,
7 compare-to ALU.cmp,
8 output-list x%B1.16.1 y%B1.16.1 zx nx zy ny f no out zr
9
10 set x %B0000000000000000, // x = 0
11 set y %B1111111111111111; // y = -1
12
13 // Compute 0
14 set zx 1,
15 set nx 0,
16 set zy 1,
17 set ny 0,
18 set f 1,
19 set no 0,
20 eval,
21 output;
22
23 // Compute 1
24 set zx 1,
25 set nx 1,
26 set zy 1,
27 set ny 1,
28 set f 1,
29 set no 1,
30 eval,
31 output;
32
33 // Compute -1
34 set zx 1,
35 set ny 1

```

HDL code: No syntax errors

Conclusion

This assignment involved the implementation of fundamental digital logic circuits using HDL in the Nand2Tetris Hardware Simulator. Through Project 1 and Project 2, various combinational components such as logic gates, multiplexers, demultiplexers, adders, and the ALU were designed and tested successfully. The assignment helped in understanding the basics of digital system design and hardware abstraction.